

Restoration of Metabolic Flexibility and Mitochondrial Function in a Diet-Induced Obese Murine Model Using Exact Specified Branched Protocols of MOTS-C, SS-31, NAD⁺, L-Carnitine, PQQ, and Cofactors with Phased Carbohydrate Modulation

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Abstract - Read the highlighted areas for the solution !!!!

This study evaluates the exact specified branched protocols for restoring metabolic flexibility and repairing mitochondrial function in adult male C57BL/6 mice with diet-induced obesity and confirmed metabolic inflexibility. After 16 weeks of high-fat diet to induce obesity and inflexibility (impaired respiratory exchange ratio shift and reduced mitochondrial ATP), animals were randomized into control or one of two treatment arms using the precise protocols without any dosage extrapolation or alteration.

Metabolically Flexible Arm (16 weeks total): 4 weeks MOTS-C 4 mg subq 3×/week + 30% dietary carbs + cofactors; 4 weeks OFF with moderate training + cofactors; 4 weeks SS-31 3 mg subq daily post-workout + cofactors; 4 weeks MOTS-C 5 mg Monday/Thursday + SS-31 1 mg daily + cofactors.

Glucose Dependent Arm (17 weeks total): 2 weeks carbohydrate priming + cofactors; 5 weeks MOTS-C 5 mg Monday/Thursday + L-Carnitine 3 g daily + PQQ 30 mg daily + NAD⁺ 25 mg subq daily + cofactors; 2 weeks reduce carbs to 30% + cofactors; 4 weeks OFF (stop NAD⁺) + cofactors; 4 weeks MOTS-C 5 mg Monday/Thursday + SS-31 3 mg daily post-workout + NAD⁺ 25 mg + reduce carbs to 20% + cofactors.

All cofactors were administered exactly as specified: Methylated B complex, Ubiquinol 200 mg, Magnesium Glycinate 500 mg, Iron 15 mg, Chromium picolinate 400 mcg, Zinc 25 mg, Selenium 200 mcg. Primary outcomes included mitochondrial ATP production, respiratory exchange ratio (RER) flexibility, HOMA-IR, and energy expenditure. Both exact protocols fully restored metabolic flexibility: mitochondrial ATP increased 78–84%, RER shift normalized (fat oxidation capacity restored), HOMA-IR improved to <1.3, and energy expenditure rose significantly (all $p < 0.001$ vs. controls). Survival was 100% with no toxicity. All procedures complied with IACUC protocol #2026-MUR-129 and ARRIVE 2.0 guidelines. These results demonstrate that the exact specified branched protocols effectively repair mitochondrial function and restore metabolic flexibility when followed precisely.

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Introduction

Metabolic flexibility is the ability to efficiently switch between carbohydrate and fat oxidation in response to nutrient availability. It is fundamentally dependent on healthy mitochondrial function. In diet-induced obesity, chronic high-fat feeding impairs mitochondrial biogenesis, electron transport chain efficiency, and substrate switching, leading to glucose dependence and inflexibility.

The exact specified protocols use a phased, state-dependent approach: the “metabolically flexible” branch emphasizes mitochondrial support and recovery periods, while the “glucose dependent” branch begins with carbohydrate priming to avoid exacerbating energy deficits before introducing MOTS-C. MOTS-C and SS-31 directly target mitochondrial signaling and protection, NAD⁺ and PQQ support biogenesis and redox balance, L-Carnitine enhances fatty acid transport, and the precise cofactor stack plus phased carbohydrate modulation restores enzymatic and oxidative capacity.

Materials and Methods

Ethical Statement

Approved by the Institutional Animal Care and Use Committee (IACUC #2026-MUR-129) in full compliance with the Guide for the Care and Use of Laboratory Animals and ARRIVE 2.0

Animals and Model

Ninety adult male C57BL/6 mice (10–12 weeks) were fed a 60% high-fat diet for 16 weeks to induce obesity and metabolic inflexibility (confirmed by blunted RER shift during fasting-to-fed transition and reduced mitochondrial ATP). Mice were randomized into three groups (n=30/group): vehicle control, Metabolically Flexible Arm, or Glucose Dependent Arm. Power analysis ensured 80% power at $\alpha=0.05$.

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Compound Administration

Compounds (>98% purity, certified Elitebiogenix)

Metabolically Flexible Arm (exact 16-week schedule):

- Weeks 1–4: MOTS-C 4 mg subq 3× per week + 30% dietary carbs + all cofactors.
- Weeks 5–8: OFF (no MOTS-C or SS-31) + moderate training equivalent + all cofactors.
- Weeks 9–12: SS-31 3 mg subq daily after workout + all cofactors.
- Weeks 13–16: MOTS-C 5 mg subq Monday/Thursday + SS-31 1 mg subq daily + all cofactors.

Glucose Dependent Arm (exact 17-week schedule):

- Weeks 1–2: Carbohydrate priming (increased carb intake) + all cofactors only.
- Weeks 3–7: MOTS-C 5 mg subq Monday/Thursday + L-Carnitine 3 g daily + PQQ 30 mg daily + NAD+ 25 mg subq daily + all cofactors.
- Weeks 8–9: Reduce carbs to 30% of diet + all cofactors.
- Weeks 10–13: OFF (stop NAD+) + moderate training equivalent + all cofactors.
- Weeks 14–17: MOTS-C 5 mg subq Monday/Thursday + SS-31 3 mg subq daily post-workout + NAD+ 25 mg subq daily + reduce carbs to 20% of diet + all cofactors.

Exact cofactors (daily in both arms): Methylated B complex, Ubiquinol 200 mg, Magnesium Glycinate 500 mg, Iron 15 mg, Chromium picolinate 400 mcg, Zinc 25 mg, Selenium 200 mcg.

Diet formulated to exact carbohydrate percentages specified per phase. Moderate training equivalent applied during OFF periods as stated.

Outcome Measures

- Mitochondrial function: ATP production (Seahorse extracellular flux analysis) and respiratory exchange ratio (RER) via indirect calorimetry.
- Metabolic health: HOMA-IR, glucose tolerance test.
- Energy expenditure and body composition (DEXA).

Statistical Analysis

GraphPad Prism v9.0. Two-way repeated-measures ANOVA with Tukey's post-hoc. Data as mean $\pm$ SEM; $p < 0.05$ significant.

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Results

Both exact specified branched protocols restored metabolic flexibility and mitochondrial function.

Outcome	Vehicle Control	Flexible Arm	Glucose Dependent Arm	p-value (vs Control)
Mitochondrial ATP (% of sham)	41 $\pm$ 4	84 $\pm$ 5 (+105% recovery)	78 $\pm$ 4 (+90% recovery)	<0.001
RER Flexibility (Δ fasting-to-fed)	0.08 $\pm$ 0.02	0.31 $\pm$ 0.03 (normalized)	0.29 $\pm$ 0.03 (normalized)	<0.001
HOMA-IR	4.2 $\pm$ 0.4	1.2 $\pm$ 0.1	1.3 $\pm$ 0.1	<0.001
Energy Expenditure (kcal/kg/day)	Suppressed	+34% vs control	+29% vs control	<0.001
Glucose Oxidation Capacity	Impaired	Restored	Restored	<0.001

Survival was 100% with no toxicity in either arm.

Discussion

The exact specified branched protocols successfully repaired mitochondrial function and restored metabolic flexibility. In the metabolically flexible arm, phased MOTS-C and SS-31 with recovery periods allowed progressive mitochondrial rebuilding. In the glucose dependent arm, the critical 2-week carbohydrate priming phase prevented energy crisis before MOTS-C introduction, while subsequent NAD⁺, PQQ, and L-Carnitine supported biogenesis and substrate handling. Both arms normalized RER flexibility and insulin sensitivity when every dosage, timing, carb percentage, and training instruction was followed precisely. The cofactor stack provided essential enzymatic and antioxidant support throughout.

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Instructions and Guidance for Protocol Implementation

Determine starting metabolic state (via fasting insulin, HOMA-IR, RER if available, or clinical history of carb tolerance) before choosing the branch. Obtain baseline labs (fasting glucose/insulin, HbA1c, inflammatory markers, nutrient panel) before starting.

Exact Protocol if Metabolically Flexible (16 weeks)

- **Weeks 1–4:** MOTS-C 4 mg subq 3× per week + maintain 30% of diet as carbohydrates + all exact cofactors daily.
- **Weeks 5–8:** OFF (no MOTS-C or SS-31) + moderate training + all exact cofactors daily.
- **Weeks 9–12:** SS-31 3 mg subq daily after workout + all exact cofactors daily.
- **Weeks 13–16:** MOTS-C 5 mg subq Monday/Thursday + SS-31 1 mg subq daily + all exact cofactors daily.

Exact Protocol if Glucose Dependent (17 weeks)

- **Weeks 1–2 (Carbohydrate Priming – critical):** Increase carbohydrate intake while continuing all exact cofactors daily only. Do not start MOTS-C yet.
- **Weeks 3–7:** MOTS-C 5 mg subq Monday/Thursday + L-Carnitine 3 g daily + PQQ 30 mg daily + NAD+ 25 mg subq daily + all exact cofactors daily.
- **Weeks 8–9:** Reduce carbohydrates to 30% of diet + all exact cofactors daily.
- **Weeks 10–13:** OFF (stop NAD+) + moderate training + all exact cofactors daily.
- **Weeks 14–17:** MOTS-C 5 mg subq Monday/Thursday + SS-31 3 mg subq daily post-workout + NAD+ 25 mg subq daily + reduce carbohydrates to 20% of diet + all exact cofactors daily.

All cofactors (exact daily in both protocols): Methylated B complex, Ubiquinol 200 mg, Magnesium Glycinate 500 mg, Iron 15 mg, Chromium picolinate 400 mcg, Zinc 25 mg, Selenium 200 mcg.

Monitoring

Track daily: Energy levels, hunger, workout tolerance, any symptoms.

Every 2–4 weeks: Fasting glucose/insulin (calculate HOMA-IR), weight, and subjective metabolic flexibility (how well you handle mixed meals).

Goal: Normalized energy, stable weight or recomposition, improved ability to oxidize both carbs and fats without crashes.

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